STOCHASTIC METHODS IN WATER RESOURCES

Unit 6: Stochastic modelling and prediction
Lecture 19: Stochastic Differential Equations in Hydrology

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Motivation in Hydrology

Hydrological systems are subject to:

- Random rainfall variability
- Subsurface heterogeneity
- Measurement noise and model uncertainty

Deterministic ODEs fail to represent these effects.
SDEs explicitly incorporate randomness:

$$dX_t = f(X_t, t)\,dt + g(X_t, t)\,dW_t$$

where W_t is a Wiener process.
A **Wiener Process (Brownian Motion)** $W(t)$ is a stochastic process defined by:

- $W(0) = 0$
- Continuous paths: $W(t)$ is almost surely continuous.
- Independent increments: $W(t) - W(s)$ independent of past for $t > s$.
- Gaussian increments:
$$W(t) - W(s) \sim \mathcal{N}(0,\, t - s)$$

Deterministic ODE vs Stochastic SDE

Deterministic model

$$\frac{dS}{dt} = I(t) - Q(S)$$

Unique trajectory for given initial condition.

Stochastic model

$$dS = (I - kS)dt + \sigma dW_t$$

Produces ensemble of possible trajectories.

- ODE: no uncertainty propagation
- SDE: state variance evolves in time

General Form of SDE

Hydrological state variable $X(t)$ (flow, storage, soil moisture):

$$dX(t) = a(X,t)dt + b(X,t)dW_t$$

Components:

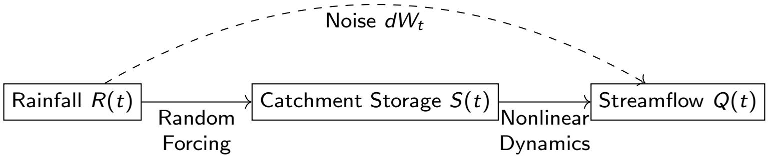
- Drift $a(X,t)$: physical dynamics
- Diffusion $b(X,t)$: stochastic forcing
- $dW(t)$: infinitesimal increment of Wiener process

Mean evolution:

$$\mathbb{E}[X(t)] \neq X_{det}(t)$$

which means that the average (mean)($\mathbb{E}[.]$) behavior of a stochastic system ($X(t)$) is not the same as the solution of the corresponding deterministic system $X_{det}(t)$

Conceptual Diagram of SDE



Compact conceptualization of stochastic rainfall-runoff.

Itô Interpretation

Increment form:

$$\Delta X = a(X,t)\Delta t + b(X,t)\Delta W$$

Properties of Wiener process:

$$\mathbb{E}[dW_t] = 0, \quad Var(dW_t) = dt$$

Important result:

$$(dW_t)^2 = dt$$

This alters chain rule (Itô lemma).

Itô Calculus in Rainfall–Runoff and Groundwater Systems

Hydrological Context

Let $X(t)$ represent a hydrological state variable such as:

- River discharge $Q(t)$ (rainfall–runoff)
- Groundwater head $h(t)$ (aquifer storage dynamics)

These variables are influenced by uncertain forcing:

- Random rainfall events
- Spatial heterogeneity of soil and aquifer properties
- Climate variability

Stochastic Hydrological Model (Itô SDE)

$$dX(t) = f(X,t)\,dt + g(X,t)\,dW(t)$$

where:

- $f(X,t)$ represents deterministic hydrological processes (infiltration, recharge, drainage),
- $g(X,t)$ controls the intensity of random forcing,
- $W(t)$ is a Wiener process modeling rainfall uncertainty.

Key Itô Property

$$(dW)^2 = dt$$

This implies that randomness modifies the physical evolution of the system, introducing an additional drift component.

Physical Interpretation

- Random rainfall does not only cause fluctuations — it alters the mean hydrograph.
- Groundwater recharge becomes statistically shifted under stochastic forcing.
- Flood peaks and storage dynamics deviate from deterministic predictions.

Ignoring stochastic forcing can lead to biased flood risk estimation and misrepresentation of long-term groundwater behavior.

Euler–Maruyama Scheme

Numerical integration for SDE:

$$X_{n+1} = X_n + a(X_n, t_n)\Delta t + b(X_n, t_n)\sqrt{\Delta t}\, Z_n$$

where:

- $Z_n \sim \mathcal{N}(0, 1)$
- Monte Carlo simulation produces ensemble

Used in:

- Flood routing uncertainty
- Basin storage estimation

Example: Stochastic Linear Reservoir

Model storage $S(t)$:

$$dS = (R - kS)dt + \sigma dW_t$$

- $S(t)$: Water storage at time t (soil moisture, reservoir storage, or groundwater storage)
- dS : Infinitesimal change in storage over time interval dt
- R : Mean recharge or effective rainfall input (average inflow to the system)
- k : Recession or drainage coefficient (controls how fast water leaves the system)
- kS : Storage-dependent outflow (baseflow, percolation, evapotranspiration losses)
- σ : Noise intensity (magnitude of rainfall or recharge variability)
- W_t : Wiener process (Brownian motion) representing random rainfall or climate fluctuations
- dW_t : Random perturbation applied to storage

Discretized:

$$S_{n+1} = S_n + (R - kS_n)\Delta t + \sigma\sqrt{\Delta t}Z_n$$

Streamflow:

$$Q_n = kS_n$$

Represents rainfall-runoff with noise.

Worked Numerical Example

Let:

- $R = 10$ mm/h
- $k = 0.2$ h^{-1}
- $\sigma = 1.5$
- $\Delta t = 1$ h, $S_0 = 20$
- $Z_0 = 0.4$

Step 1:

$$S_1 = 20 + (10 - 0.2 \times 20)(1) + 1.5(1)(0.4)$$

$$S_1 = 20 + (10 - 4) + 0.6 = 26.6$$

Then $Q_1 = 0.2 \times 26.6 = 5.32$ mm/h

Hydrological Applications

- Flood wave propagation with uncertainty
- Groundwater head fluctuation modeling
- Soil moisture stochastic balance
- Climate-driven runoff variability
- Real-time forecasting with noise-aware models

SDEs provide physically realistic uncertainty dynamics.

Advantages and Limitations

Advantages

- Explicit uncertainty representation
- Realistic ensemble generation
- Suitable for data assimilation

Limitations

- Requires stochastic calculus
- Parameter estimation more complex
- Computationally intensive

Summary

- SDEs extend hydrological models by including randomness
- Governed by drift + diffusion components
- Solved numerically with Euler–Maruyama
- Useful for realistic flood and runoff modeling
- Foundation for Kalman filtering and data assimilation

SDEs bridge physics and uncertainty in modern hydrology.